quantization_methods_int8 (Initial Draft)

Domain: Machine Learning | Document ID: quantization_methods_int8_v1
Specification Standard: IEEE Capstone Evaluation Corpus

Executive Overview:

Post-training quantization (PTQ) and quantization-aware training (QAT) mapping 32-bit floating point weights to INT8 representations.

Methodological Verification & System Lineage:

This specification is catalogued in IntentCloud for intent-aware hybrid retrieval.
Text chunks are indexed into 384-dimensional dense semantic vectors and 1,000,003-dimensional sparse feature-hashed token projections.
Candidate pools are unified using scale-invariant Reciprocal Rank Fusion (k=60).
Explainable citation attribution is extracted using cross-encoder cross-attention.